

EBN111 Practical 2 Instructions

(To be completed during the practical session)

2013

Print this document and bring it to the practical. You need to complete this during the practical session.

After completing this document, log on to ClickUP and fill in the answers in **Practical 2 Online Evaluation**.

NB:

- Use a full stop for denoting a decimal point. **DO NOT** use a comma.
- Use 2 decimal digits for all answers.
- Express your answers in the specified units.
- **DO NOT** supply units on ClickUp.
- The laboratory instruments are sensitive and expensive, use with care.

Introduction

The aim of this practical is to introduce you to the basic equipment used when building electric circuits. You will build a few simple electric circuits using simple passive electronic components in order to test a number of basic electronic concepts.

It is important that you understand the theory before building and testing your circuits.

Make sure that you have read through the **Practical 2 Laboratory Guide** before starting with this practical.

You will need to use the components in the box that you have received.

EBN111 Practical 2

Question 1

[4]

Objective

This question introduces the power supply and the multimeter. The goal here is to realize that a measured value and the theoretical value is not always 100% the same.

Instructions

1. Before you switch on the power supply, make sure all the voltage knobs are turned down to the minimum.
2. Connect the power supply directly to the multimeter as shown in Figure 1. Remember that you are measuring DC voltage, so make sure the multimeter is set to measure this. The dashed lines denote cables found in your box of components.
3. After setting up the equipment, turn on the power supply.
4. Complete Table 1.

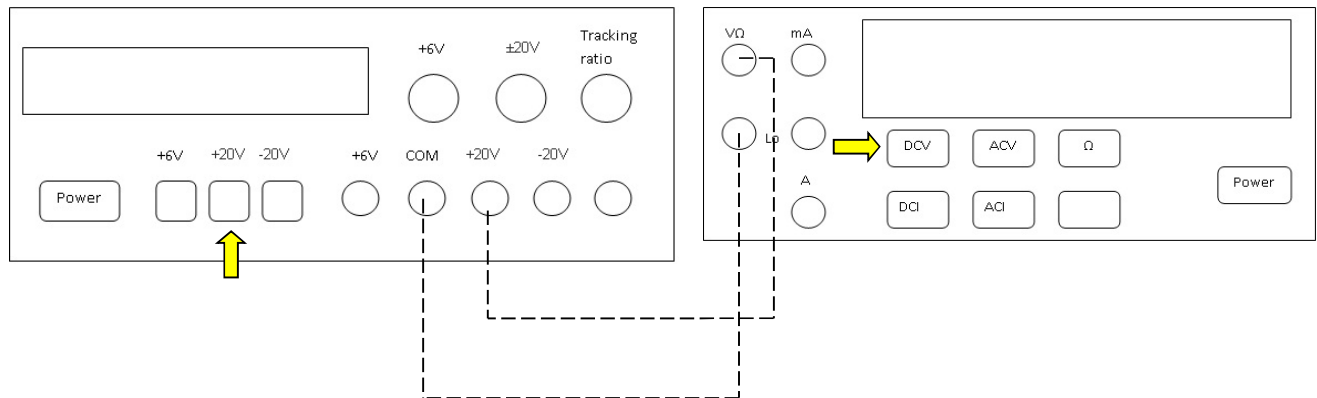


Figure 1. Connecting the multimeter to the power supply.

Voltage supplied [V] (Set the power supply to these values)	Voltage measured [V] (Read off from the multimeter)
1	
2	
10	
20	

Table 1. Voltage supplied and voltage measured.

Question 2

[3]

Objective

This question involves measuring resistance using the multimeter.

Instructions

Find a 12 Ω , 1 k Ω and 10 k Ω resistor and measure the resistances using the multimeter. Complete Table 2.

Defined resistance [Ω] (From the colour code on the resistor)	Measured resistance (Read off multimeter)
12	Ω
1000	k Ω
10 000	k Ω

Table 2. Defined resistance and measured resistance.

Question 3

[7]

Objective

In this question you will monitor a signal generated by the function generator using the oscilloscope.

Instructions

Connect the equipment as shown in Figure 2 and follow the steps below.

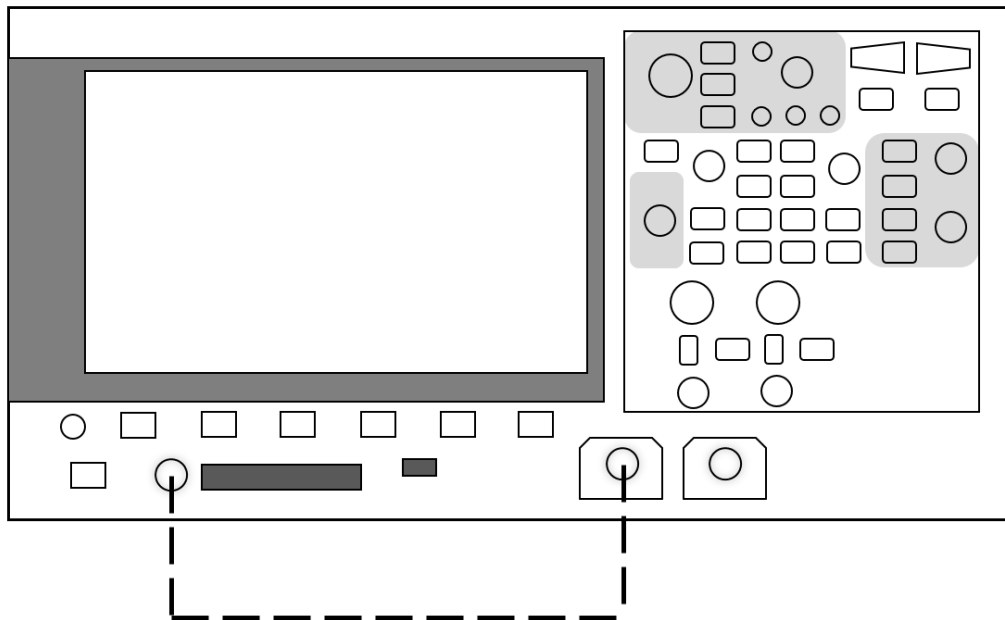


Figure 2. Connecting the function generator to the oscilloscope.

1. Connect the **Gen Out** to the channel 1 of the oscilloscope (the yellow channel). Use the BNC cables and connect the crocodile clip ends black to black and red to red.
2. Switch on the oscilloscope.
3. Push the **Wave Gen** button on the oscilloscope.
4. Select the **Waveform** to be a sine wave with frequency 2 kHz, 1 Vpp Amplitude and 0 V offset.
5. Now scale your graph until you clearly see the sine wave on the screen. Use the **Horizontal** and **Vertical** knobs to do this (push to zero).
6. Push the **Meas** button on your scope.
7. Add the following measurements: Frequency, period and rise-time. Do this by selecting the **Type** and then adding the measurement
8. Press the **Run/Stop** button.
9. Complete Table 3.

Measured frequency [kHz]	Measured period [μ s]	Measured rise-time [μ s]

Table 3. Values measured using the oscilloscope.

Question 4

[3]

Objective

In this question you will test the principle of voltage division. Figure 3 illustrates the concept. You will solve the circuit theoretically and then measure the value by building the actual circuit.

Instructions

For $V_G = 5$ V, $R_1 = 1$ k Ω and $R_2 = 10$ k Ω . Calculate V_2 theoretically (using the equation for voltage division).

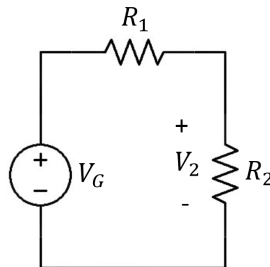


Figure 3. Circuit diagram for Question 4.

Now build the circuit and measure the voltage V_2 , refer to Figure 4. Set your power supply to the voltage of V_G (5 V). Complete Table 4.

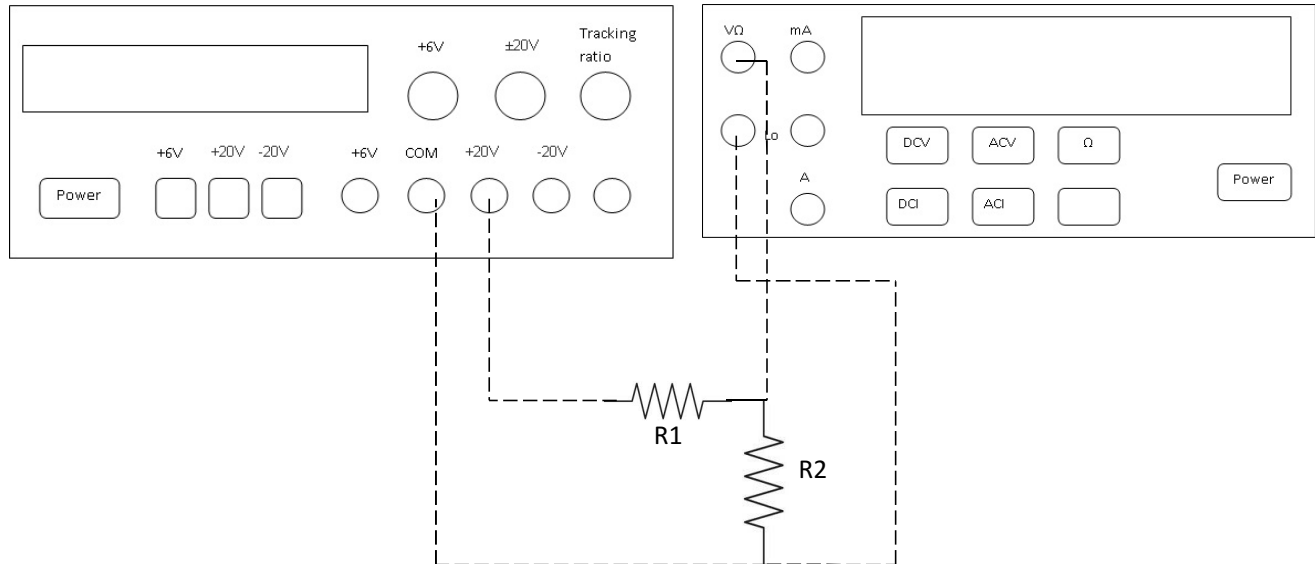


Figure 4. Wiring diagram for Figure 3.

Theoretical value for V_2 [V]	Measured value for V_2 [V]

Table 4. Theoretical and measured values for the voltage V_2 .

Question 5

[6]

Objective
In this question you will test the concept of current division. Figure 5 illustrates the concept. You will solve the circuit theoretically and then measure the values by building the actual circuit.

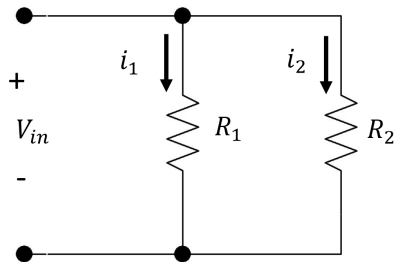


Figure 5. Circuit diagram for Question 5.

Instructions
For $R_1 = 1\text{ k}\Omega$ and $R_2 = 10\text{ k}\Omega$ and $V_{in} = 5\text{ V}$, determine the theoretical values of i_1 and i_2 .

Theoretical value for i_1 [mA]	Theoretical value for i_2 [μA]

Table 5. Theoretical values for i_1 and i_2 .

Now build the circuit and measure the actual currents i_1 and i_2 . Figure 6 shows the circuit you must build to measure i_2 . Complete Table 6.

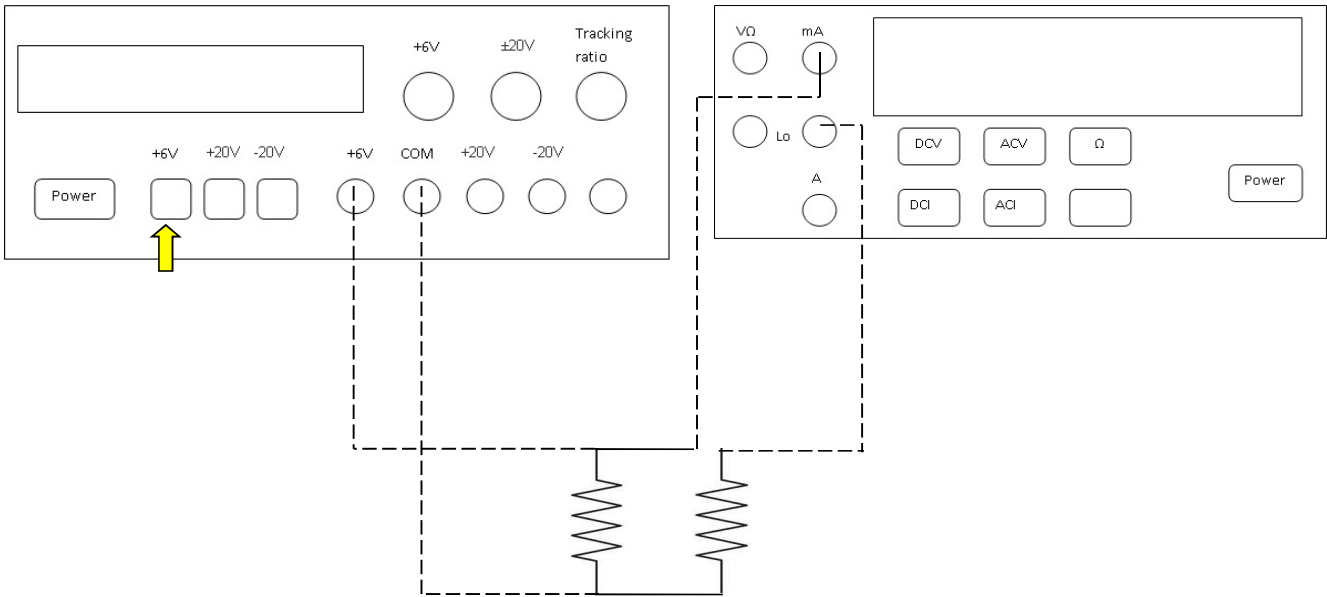


Figure 6. Wiring diagram for Figure 5 where i_2 is measured.

Measured value for i_1 [mA]	Measured value for i_2 [μA]

Table 6. Measured values for i_1 and i_2 .

Question 6

[9]

Instructions
Build the circuit and measure the current i_x , the node voltage V_1 and the voltage V_o shown in Figure 7. Complete Table 7.

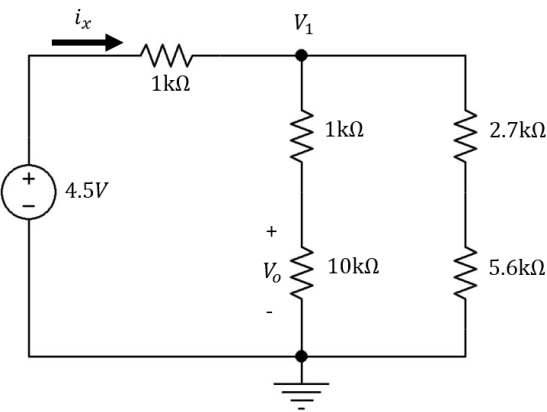


Figure 7. Circuit diagram for Question 6.

i_x [μ A]	V_1 [V]	V_o [V]

Table 7. Measured values for Question 6.

Question 7

[4]

Instructions
Measure R_{eq} using the multimeter. Complete Table 8.

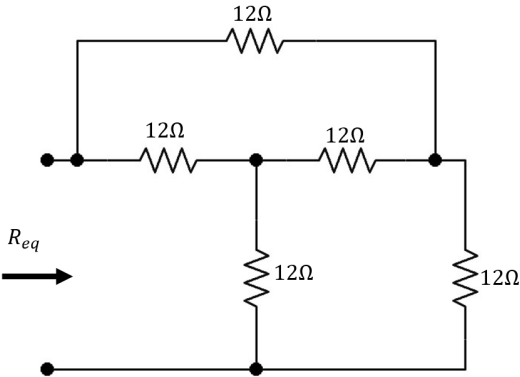


Figure 8. Circuit diagram for Question 7.

R_{eq} [Ω]

Table 8. Equivalent resistance of the circuit in Figure 8.

Question 8

[8]

Instructions

Build the circuit in Figure 9 and measure the values of i_x and V_o . Complete Table 9.

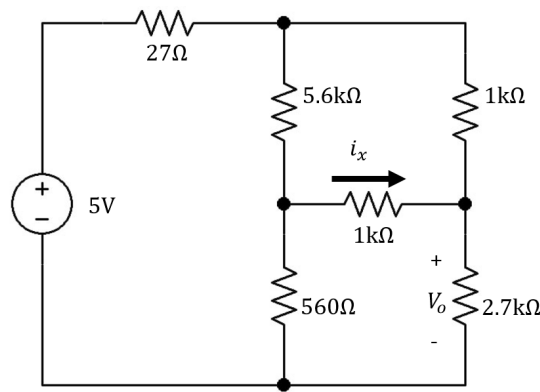


Figure 9. Circuit diagram for Question 8.

i_x [mA]	V_o [V]

Table 9. Measured values for the circuit in Figure 9.

Now enter all your answers on ClickUp in **Practical 2 Online Evaluation**.